MTH 331(A) Dr. James Ashe

9.1 Worksheet Wednesday 21st August, 2013 **Name:**

1. Find the sequence generated by the formula $a_n = \sin\left({}^{n\pi}/_2\right)/n$, for $n = 1, 2, 3, \ldots$.

2. Find the sequence generated by the recurrence relation $a_0 = 1$, $a_{n+1} = {}^{a_n^2}/_2$ for $n = 1, 2, 3, \ldots$.

3. Suppose you deposit \$100 in a bank account. Once per year the balance is increased by 5% due to interest and you also deposit \$50. The sequence $\{B_n\}$ that gives the balance in the account after n years satisfies the recurrence relation $B_0 = \$100$ and ________________.

4. Write out the terms of $\displaystyle\sum_{n=1}^{\infty} \frac{n-4}{n}$.